



CHAPTER 1

PROBABILITY

PAGE 1 • COMPLETE THEORY + ALL BASIC FORMULAS

1. BASIC TERMS

- Random experiment: an experiment whose exact outcome cannot be predicted in advance.
- Trial: one performance of a random experiment.
- Outcome: one possible result of a trial.
- Event (E): a collection of one or more outcomes.
- Sample space (S): the set of all possible outcomes.

- Favourable outcomes n(E): outcomes

2. EVENT TYPES

- Impossible event: event that cannot occur. $P(\emptyset)=0$.
- Sure event: event that always occurs. $P(S)=1$.
- Simple event: event containing exactly one outcome.
- Compound event: event containing more than one outcome.
- Complementary event: E' or not E.
- Mutually exclusive events: events that cannot occur together.
- Equally likely outcomes: outcomes having the same chance of occurring.

3. CORE FORMULA

When all outcomes are equally likely:

$$P(E) = n(E) / n(S)$$

where $n(E)$ = favourable outcomes and $n(S)$ = total outcomes.

$$P(\text{not } E) = 1 - P(E)$$

4. RANGE AND EXTREME EVENTS

$$0 \leq P(E) \leq 1$$

- If $P(E)=0$, E is impossible.
- If $P(E)=1$, E is sure/certain.
- If $0 < P(E) < 1$, E is possible.
- Probability can never be negative.
- Probability can never be greater than 1.
- More favourable outcomes usually means higher probability.

5. COMPLEMENT RULE

$$P(E') = 1 - P(E)$$

$$P(E) + P(E') = 1$$

- Use complement when direct counting is long.
- 'Not getting a head' is the complement of 'getting a head'.
- 'Not an ace' is the complement of 'getting an ace'.
- For any event E, E and E' are mutually exclusive and exhaustive.

6. SAMPLE SPACE COUNTING

- One coin: $S=\{H,T\}$; $n(S)=2$.
- Two coins: $S=\{HH, HT, TH, TT\}$; $n(S)=4$.
- Three coins: $n(S)=2^3=8$.
- One die: $S=\{1,2,3,4,5,6\}$; $n(S)=6$.
- Two dice: $n(S)=6 \times 6=36$.
- Deck of cards: $n(S)=52$.
- Number cards 1 to n: $n(S)=n$.

7. ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive

$$P(A \cup B) = P(A) + P(B)$$

- Use u for 'or'.
- Use n for 'and'.
- CBSE basic chapter mainly uses simple counting, but this rule prevents confusion.

8. MULTIPLICATION IDEA

$$\text{Total cases} = \text{choices}_1 \times \text{choices}_2 \times \dots$$

- Coin tossed 3 times: $2 \times 2 \times 2 = 8$ outcomes.
- Two dice: $6 \times 6 = 36$ outcomes.
- Choosing one card: 52 outcomes.
- For combined independent choices, multiply counts.
- Do not multiply probabilities unless the question needs combined events.

9. QUICK DECISION TREE

1

Question asks direct chance?
List total outcomes and favourable outcomes.

2

Question says 'not'?
Use complement: $1 - P(E)$.

3

Question says 'at least one'?
Use complement of none.

4

Question uses coin/dice/cards?
Write sample space count first.

5

Final answer?
Simplify fraction and check $0 \leq P \leq 1$.



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PAGE 2 • COINS, DICE, CARDS + STANDARD OUTCOMES

1. COINS

- One coin: $S = \{H, T\}$; $n(S) = 2$.
- Two coins: $S = \{HH, HT, TH, TT\}$; $n(S) = 4$.
- Three coins: $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$; $n(S) = 8$.
- n coins: total outcomes = 2^n .
- Exactly r heads in n tosses: favourable outcomes = $C(n, r)$.
- At least one head = $1 - P(\text{no head})$.

$$P(\text{exactly } r \text{ heads}) = C(n, r) / 2^n$$

2. DICE

- One die: $S = \{1, 2, 3, 4, 5, 6\}$; $n(S) = 6$.
- Two dice: total outcomes = $6 \times 6 = 36$ ordered pairs.
- Even number on one die: $\{2, 4, 6\}$; probability = $3/6 = 1/2$.
- Prime number on one die: $\{2, 3, 5\}$; probability = $3/6 = 1/2$.
- Number greater than 4: $\{5, 6\}$; probability = $2/6 = 1/3$.
- Sum in two dice: count ordered pairs carefully.

3. TWO DICE SUM TABLE

Sum	2	3	4	5	6	7	8	9	10	11	12
Count	1	2	3	4	5	6	5	4	3	2	1
Prob.	1/36	2/36	3/36	4/36	5/36	6/36	5/36	4/36	3/36	2/36	1/36

Most common mistake: for two dice, (2,5) and (5,2) are different ordered outcomes. So count ordered pairs, not just combinations.

- Sum 2: $(1, 1) \rightarrow 1$ way
- Sum 7: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1) \rightarrow 6$ ways
- Sum 12: $(6, 6) \rightarrow 1$ way
- Probability of sum $\geq 10 = (3+2+1)/36 = 6/36 = 1/6$

4. PLAYING CARDS

- Standard deck: 52 cards.
- 4 suits: Hearts, Diamonds, Clubs, Spades.
- Red cards: Hearts + Diamonds = 26.
- Black cards: Clubs + Spades = 26.
- Face cards: J, Q, K in each suit = 12.
- Aces: 4.
- Kings: 4, Queens: 4, Jacks: 4.
- Number cards 2 to 10 in each suit = $9 \times 4 = 36$.

5. CARD PROBABILITY FAST TABLE

Getting an ace	$4/52 = 1/13$
Getting a king	$4/52 = 1/13$
Getting a queen	$4/52 = 1/13$
Getting a jack	$4/52 = 1/13$
Getting a red card	$26/52 = 1/2$
Getting a black card	$26/52 = 1/2$
Getting a face card	$12/52 = 3/13$
Getting a heart	$13/52 = 1/4$
Not getting an ace	$48/52 = 12/13$



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PAGE 3 • BOARD/JEE-FOUNDATION STYLE QUESTION TYPES

1. UNIVERSAL SOLVING METHOD

Step 1

Identify the experiment.

Step 2

Write/count the sample space S .

Step 3

Define the event E exactly.

Step 4

Count favourable outcomes $n(E)$.

Step 5

Apply $P(E) = n(E)/n(S)$.

Step 6

Simplify and check $0 \leq P \leq 1$.

2. 'AT LEAST ONE' TYPE

Use complement. This is faster and avoids missing cases.

$$P(\text{at least one success}) = 1 - P(\text{no success})$$

- At least one head in 2 tosses: $1 - P(TT) = 1 - 1/4 = 3/4$.
- At least one head in 3 tosses: $1 - P(TTT) = 1 - 1/8 = 7/8$.
- At least one six in one die throw: same as getting six = $1/6$.
- At least one six in two dice: $1 - P(\text{no six on both dice}) = 1 - (5/6)^2 = 11/36$.

3. 'NOT' TYPE

$$P(\text{not } E) = 1 - P(E)$$

- Not getting head in one toss: $1 - 1/2 = 1/2$.
- Not getting 5 on a die: $1 - 1/6 = 5/6$.
- Not getting an ace from cards: $1 - 4/52 = 48/52 = 12/13$.
- Not getting a red card: $1 - 26/52 = 1/2$.
- Not getting a face card: $1 - 12/52 = 40/52 = 10/13$.

4. 'OR' TYPE

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

- For mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B)$.
- Die: $P(2 \text{ or } 5) = 1/6 + 1/6 = 2/6 = 1/3$.
- Card: $P(\text{king or queen}) = 4/52 + 4/52 = 8/52 = 2/13$.
- Card: $P(\text{ace or spade}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$.
- Subtract overlap once if the event can be both.

5. 'AND' / COMBINED CASES

In Class 10, most questions avoid advanced conditional probability. Count directly whenever possible.

- Two dice: $P(\text{both even}) = \text{favourable pairs} / 36 = 3 \times 3 / 36 = 1/4$.
- Two coins: $P(\text{one head and one tail}) = 2/4 = 1/2$.
- One card: $P(\text{red and king}) = 2/52 = 1/26$.
- One card: $P(\text{black and face card}) = 6/52 = 3/26$.
- Do not blindly multiply unless the events are independent and the experiment structure allows it.

6. HIGH-FREQUENCY WORKED PATTERNS

- Die: $P(\text{number less than 4}) = \{1, 2, 3\}/6 = 3/6 = 1/2$.
- Die: $P(\text{composite number}) = \{4, 6\}/6 = 2/6 = 1/3$.
- Die: $P(\text{prime number}) = \{2, 3, 5\}/6 = 3/6 = 1/2$.
- Two dice: $P(\text{sum} = 7) = 6/36 = 1/6$.
- Two dice: $P(\text{sum} > 9) = \text{sums } 10, 11, 12 = (3+2+1)/36 = 1/6$.
- Card: $P(\text{heart}) = 13/52 = 1/4$.
- Card: $P(\text{red queen}) = 2/52 = 1/26$.
- Card: $P(\text{black king}) = 2/52 = 1/26$.
- Card: $P(\text{number card}) = 36/52 = 9/13$.
- Card: $P(\text{neither red nor king}) = 1 - P(\text{red or king}) = 1 - (26+4-2)/52 = 24/52 = 6/13$.



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PAGE 4 • COMMON MISTAKES + COMPLETE RECAP

1. COMMON MISTAKES TO AVOID

- Writing probability greater than 1.
- Forgetting to simplify fractions.
- Counting two-dice outcomes as 21 instead of 36.
- Treating (2,5) and (5,2) as the same in dice questions.
- Forgetting that a deck has 52 cards, 26 red cards, 26 black cards.
- Confusing face cards with all picture/number cards.
- Using experimental probability when the question asks theoretical probability.
- Not listing sample space for small experiments.

2. CLASSICAL vs EXPERIMENTAL

Aspect	Theoretical	Experimental
Basis	Equally likely outcomes	Actual repeated trials
Formula	$n(E)/n(S)$	successful trials / total trials
Used	Before experiment	After experiment
Nature	Fixed theoretical value	May change trial to trial
Long run	Benchmark	Approaches theoretical value

3. COMPLETE FORMULA RECAP

Classical probability	$P(E) = n(E) / n(S)$	Range	$0 \leq P(E) \leq 1$
Experimental probability	$P(E) = \text{trials where E occurs} / \text{total trials}$	Mutually exclusive	$P(A \cup B) = P(A) + P(B)$
Complement	$P(E') = 1 - P(E)$	General addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Exhaustive complement	$P(E) + P(E') = 1$	n coins	Total outcomes = 2^n
Impossible event	$P(\emptyset) = 0$	Two dice	Total outcomes = 36
Sure event	$P(S) = 1$	Deck	Total outcomes = 52

4. WORD MEANING → MATH ACTION

- 'not' → complement → $1 - P(E)$.
- 'at least one' → $1 - P(\text{none})$.
- 'either A or B' → addition rule.
- 'both' → count intersection / combined favourable outcomes.
- 'equally likely' → classical probability is allowed.
- 'randomly selected' → all listed objects usually equally likely.
- 'with replacement' → total count resets each time.
- 'without replacement' → total count changes after selection.

5. EXAM PRESENTATION FORMAT

- 1 Given: Write the experiment.
- 2 **Sample space:** $S = \{\dots\}$ or $n(S) = \dots$
- 3 **Event:** Let $E = \text{required event}$.
- 4 Favourable outcomes: $n(E) = \dots$
- 5 **Formula:** $P(E) = n(E)/n(S)$.
- 6 Substitution: $P(E) = \dots / \dots$
- 7 **Final answer:** Simplified fraction.

6. FULL PRACTICE MAP

- Coins: one coin, two coins, three coins, exactly one head, at least one head, no head.
- Dice: even, odd, prime, composite, greater than, less than, multiple of, sum of two dice.
- Cards: red/black, suits, ace, king, queen, jack, face card, number card, complement card questions.